

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
9.	(a)		Indicative content to identify halides • addition of silver nitrate • chlorides give white precipitate, bromide gives cream precipitate to identify magnesium compounds • addition of sulfuric acid • magnesium compounds give solutions, barium and calcium compounds give precipitates to identify barium compounds • addition of sodium hydroxide • barium compounds give solutions, calcium and magnesium compounds give precipitates • once compound identified further reagents not added to it 5-6 marks Describes full details of all the tests and observations and how they allow all five compounds to be identified. <i>The candidate constructs a relevant, coherent and logically structured account including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i> 3-4 marks Describes most of the tests in detail including observations that allow most of the compounds to be identified. <i>The candidate constructs a coherent account including most of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary are generally sound.</i> 1-2 marks Describes some of the tests but lacks details and observations to allow identification of the compounds. <i>The candidate attempts to link at least two relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>		4	2	6		6

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	(b)			flame test (1) brick-red colour (1)	2			2		2
	(c)	(i)		$\text{Ba}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{BaCO}_3(\text{s})$		1		1		
		(ii)		to ensure that all the carbonate ions were precipitated	1			1		1
	(d)			some of the carbonates decompose / evolve CO_2 on heating (1) this will result in a lower mass being recorded (1)		1				
							1	2		2
	(e)	(i)		calcium chloride contains a lattice of oppositely charged ions (that are attracted to each other) / in the solid state the ions cannot move freely (1) in the molten state / in solution the ions are free to move and carry electricity (1)	2			2		
		(ii)		in iodine pairs of atoms are bonded covalently to form molecules which are held together by van der Waals forces (1) these forces are weak / easy to break (1)	2			2		
		(iii)		bonding in magnesium consists of a regular array of metal ions surrounded by a 'sea' of delocalised valence electrons (1) when a force is applied the layers of metal ions slide over each other forming a new shape (1)	2			2		
				Question 9 total	9	6	3	18	0	11